
3AL Test 1 2022

Time: 90 minutes

Available marks: 85

Maximum marks: 75

Instructions

1. Answer all questions in the lined answer booklets.
 2. Start each question on a new page.
 3. Make sure each answer is clearly labelled.
 4. Show all work and reasoning.
 5. This test consists of **4** questions and **2** numbered pages. Make sure your question paper is complete.
 6. If you've read the instructions draw a smiley face on the front of your booklet :)
Good luck! I'll give you an extra mark if you do it, no joke
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1. Let G be a group and let $H = Z(G)$.
 - (a) Prove that if G/H (where $H = Z(G)$) is cyclic with generator gH , then every element of G can be written in the form $g^n h$ for $n \in \mathbb{Z}$ and $h \in H = Z(G)$.
 - (b) Prove that if $G/Z(G)$ is cyclic, then G is abelian.
 - (c) Prove that if $|G : Z(G)| = 4$, then $G/Z(G) \cong K_4$.

[5 + 5 + 15 = 25 marks]

2.
 - (a) State the First Isomorphism Theorem (any version).
 - (b) Let $\varphi : G \rightarrow G'$ be an epimorphism. Suppose that B is a normal subgroup of G' and let $A = \{g \in G : \varphi(g) \in B\}$. Construct an epimorphism $\tilde{\varphi} : G \rightarrow G'/B$ and use the First Isomorphism Theorem to show $A \triangleleft G$.

[5 + 15 = 20 marks]

3. (a) State the Third Isomorphism Theorem.

(b) Let a and b be positive integers such that $a \mid b$. Determine the order of the factor group $a\mathbb{Z}/b\mathbb{Z}$.

[5 + 15 = 20 marks]

4. Consider the group $G = \mathbb{Z}_{18}$ with subgroup $H = \langle [9] \rangle$.

(a) Determine all subgroups of $\mathbb{Z}_{18}$ and draw the lattice diagram.

(b) Use the Correspondence Theorem to draw the lattice diagram for $\mathbb{Z}_{18}/H$.

[10 + 10 = 20 marks]